



DEPARTMENT OF MATHEMATICS AND COMPUTING
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

B.Tech. Second Semester End Semester Examination 3 August 2026 Time: 3hrs.
Integral Calculus and Difference Equations (MAIC 102/ MAIR 12) Max. Marks:50

Note: Attempt all questions and answer each question in sequence.

Q1.	Find the angle between the two surfaces $z = x^2 + y^2$ and $z = 2x^2 - 3y^2$ at the point of intersection (2, 1, 5).	(5M) ✓
Q2.	Find all the eigenvalues and eigenfunctions of Sturm-Liouville boundary value problem $y'' + \lambda y = 0$, $y'(0) = y'(L) = 0$.	(5M) ✓
Q3.	Evaluate $\iint (x + y)^2 dx dy$, over the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{a^2} = 1$.	(5M) ✓
Q4.	Evaluate $\iint \sqrt{\frac{(1-x^2-y^2)}{(1+x^2+y^2)}} dx dy$ over the positive quadrant of the circle $x^2 + y^2 = 1$ by changing to polar coordinate.	(5M) ✓
Q5.	Solve the difference equation $y_{n+2} - y_{n+1} + y_n = 0$, given that $y_0 = 1$ and $y_n = \frac{(1 + \sqrt{3})}{2}$	(5M) ✓
Q6.	Find the Z transform of $\cos n\theta$. Then using the scaling property of Z transform, obtain the Z-transform of $r^n \cos n\theta$, $n \geq 0$.	(5M) ✓
Q7.	Using the Z-transform, solve the following difference equation subject to the given condition : $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$ given $y_0 = y_1 = 0$.	(5M) ✓
Q8.	Find series solution of $xy'' + y' - y = 0$ about $x = 0$ using Frobenius method.	(5M) ✓
Q9.	State and verify Green's theorem in the plane for $\oint_C (3x^2 - 8y^2)dx + (4y - 6xy) dy$ where C is the boundary of the region defined by $x = 0, y = 0, x + y = 1$. Or Obtain the Fourier series for the function $f(x) = x^2$, $-\pi \leq x \leq \pi$.	(5M)
Q10.	Write a short note on the Fourier series. Illustrate your answer by taking any periodic function of your choice and determining its Fourier series. Or Find the workdone by the force field $F = x^2 i + yz j + z k$ in moving a particle along the line segment C from the point P: (1,2,2) to point Q: (3,4,2)	(5M) ✓